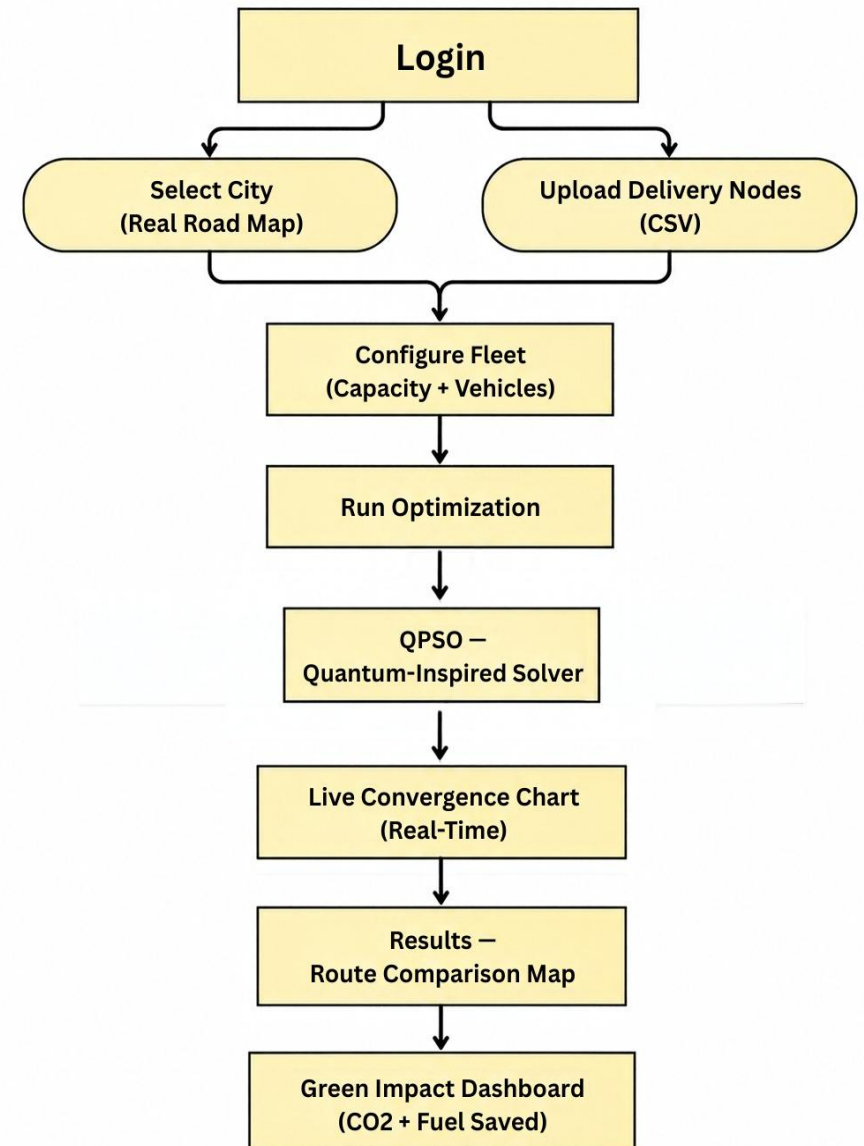


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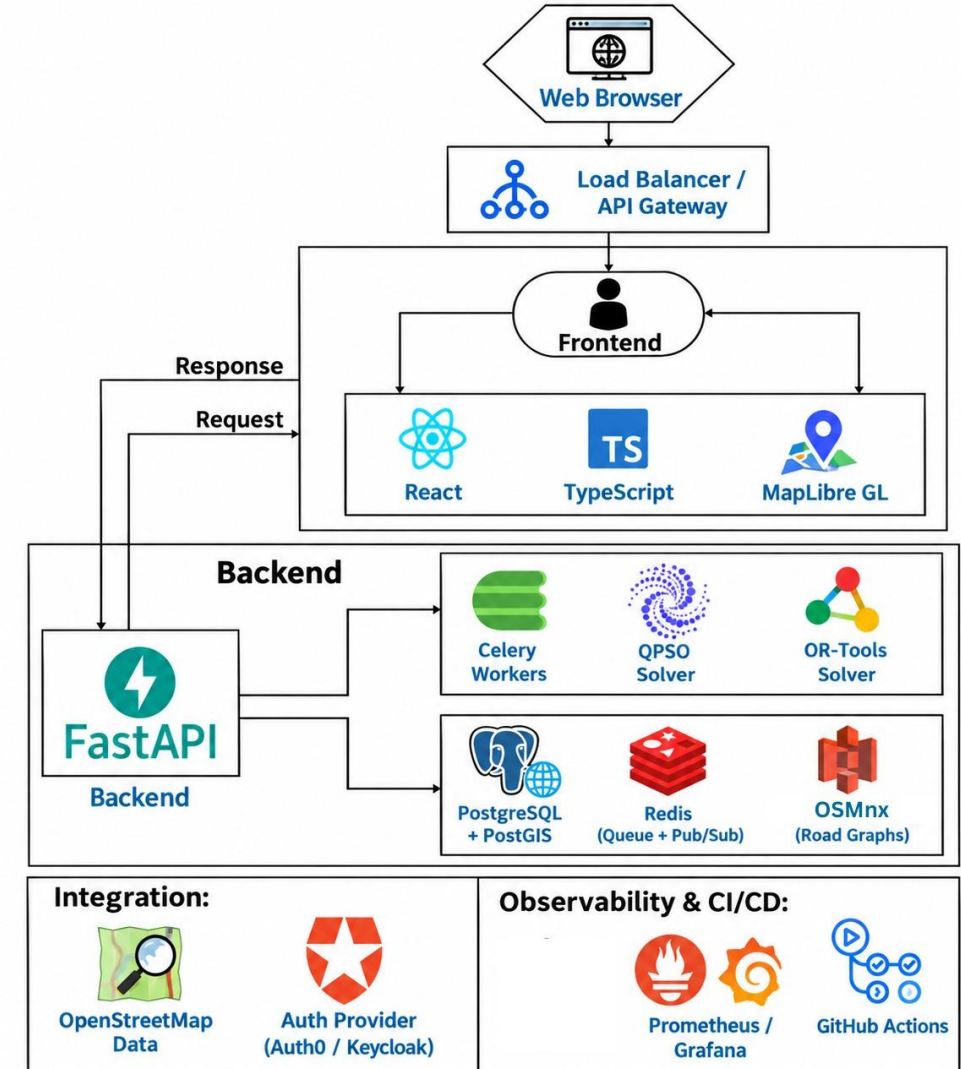
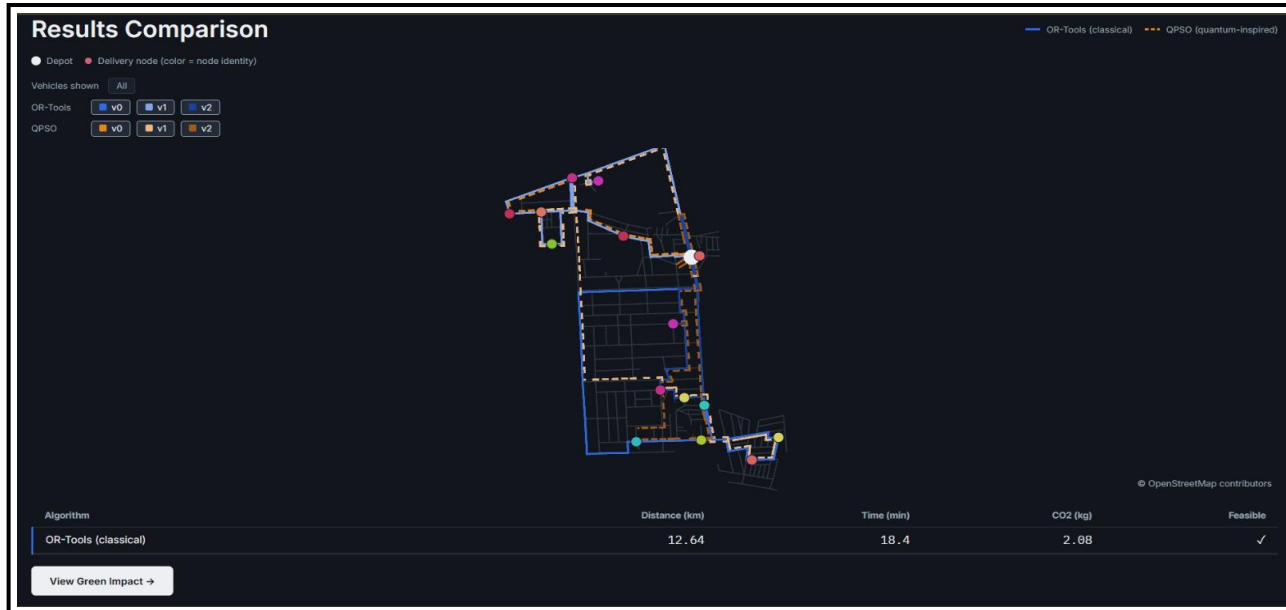
- **Problem Statement ID – SIH26137**
- **Problem Statement Title - Quantum-Inspired Intelligent Traffic Route Optimization in Transportation Systems Using Metaheuristic Optimization**
- **Theme - Transportation & Logistics**
- **PS Category - Software**
- **Team ID - 165121**
- **Team Name – AlgoNauts_LJIET**

❖ Proposed Solution:

- [Real Street Maps](#) : Loads real, live road networks for any Indian city, instantly.
- [Two Brains Compare](#) : A classical planner and our quantum-inspired planner run side by side, so you always see the smarter route.
- [Live Thinking Chart](#) : Watch the quantum planner explore thousands of options in real time.
- [Handles Real Accidents](#) : Detects a road disruption and re-routes every affected delivery automatically.
- [Green Impact Dashboard](#) : Reports the exact fuel and CO2 saved on every single plan.
- [City-Wide Load Balancing](#) : Spreads many delivery fleets across different roads so no single street gets overloaded.
- [Secure Multi-Tenant Access](#) : Every company or city team gets its own secure login and fully private data.
- [Scales With Demand](#) : Runs many optimizations at once, automatically adding more computing power as needed.



- **API & Orchestration**: FastAPI (Python, async, Pydantic-validated)
- **Routing Solvers**: Google OR-Tools (classical) + QPSO (in-house, quantum-inspired)
- **Job Execution**: Celery + Redis — horizontally scalable workers
- **Geospatial Data**: PostgreSQL + PostGIS, OSMnx-derived road graphs
- **Emission Modeling**: COPERT (EMEP/EEA formula)
- **Real-Time Streaming**: Server-Sent Events over Redis Pub/Sub
- **Frontend**: React + TypeScript + MapLibre GL
- **Platform**: Auth0 / Keycloak · Docker · GitHub Actions · Grafana



❖ Feasibility:

1. Technical

- Every component (FastAPI, Celery, PostgreSQL+PostGIS, Redis) is mature and battle-tested — low technical risk.
- Async-first design keeps solver execution and live streaming responsive under concurrent load.

2. Operational

- Deploys on managed services (RDS, Upstash, Auth0) — no infrastructure built from scratch, no Kubernetes required.
- Standard managed tiers exist for every component, so ops overhead stays low as usage grows.

3. Algorithm Core Unchanged

- Solver layer (OR-Tools, QPSO, COPERT) is already built and tested — this adds infrastructure, not algorithm risk.
- 99+ automated tests already cover the solver, emissions, and API layers end to end.

❖ Risks & Fixes:

1. Onboarding Load

- Risk:** Large graph ingestion can bottleneck new cities.
- Fix:** Async pre-fetch/cache queue ahead of demand.

2. QPSO Mid-Scale Gap

- Risk:** Trails classical baseline at ~15 nodes.
- Fix:** Root-caused, tracked, active fix in progress.

3. Data Isolation

- Risk:** Shared infra raises tenant-leak stakes.
- Fix:** DB row security + managed auth (Auth0).

❖ Viability:

1. Horizontally Scalable by Design

- Celery + Redis workers scale by adding replicas, not by redesigning the job pipeline.
- Multi-city onboarding is a background data job, not a code change, via automated OSMnx ingestion.

2. Cost-Efficient at Every Stage

- Open-source core (Postgres, Redis, Celery) with managed tiers that scale with usage — no proprietary licensing.
- Pay-as-you-grow infrastructure keeps early-stage costs low without a later re-architecture.

3. Proven Differentiator, Now Deployable

- COPERT-based ecological quantification and fair seeded benchmarking carry over unchanged from the validated core.
- Real, measured 7.6% CO2 reduction at city scale — a concrete number, not a projection.

❖ Impacts:

1. Cleaner Cities:

- Less fuel burned across every fleet, real city-wide change.
- Supports municipal air-quality and sustainability goals.

2. Smarter Logistics Industry:

- Fewer trucks, less fuel, faster deliveries for any adopter.
- Raises the bar for optimized routing in India.

3. Systemic Traffic Relief:

- Balances loads across fleets, easing single-road congestion
- Cuts bottlenecks from every operator picking the same route.

4. Research & Benchmarking Value:

- Our fair-comparison method can test any future algorithm
- Contributes an honest QPSO-vs-classical benchmark to the field.

❖ Benefits:

1. Economic Benefits:

- Fewer wasted kilometres cut real fuel costs for every fleet.
- Fewer trucks needed for the same delivery volume.

2. Environmental Benefits:

- Real, formula-sourced reporting on every kg of CO₂ avoided.
- Encourages lower-emission logistics practices industry-wide.

3. Operational Benefits:

- Automatically replans around accidents and road closures.
- Many companies run jobs simultaneously without slowdown.

4. Trust & Transparency:

- Every number ships with the exact formula behind it
- Fair, seeded benchmarking makes every result reproducible.

RESEARCH AND REFERENCE

1. [QPSO Algorithm](#): Sun, J, et al.

A Quantum-Behaved Particle Swarm Optimization Algorithm, MOPI
mdpi.com/2227-7390/10/22/4168

2. [Google OR-Tools](#): Classical vehicle-routing solver our fair baseline
developers.google.com/optimization/routing/vrp

3. [OSMnx](#): Boeing, G.-real road-network data from OpenStreetMap
osmnx.readthedocs.io

4. [COPERT Emissions Model](#): EMEP/EEA Air Pollutant Emission Inventory Guidebook
copert.emisia.com

5. [Server-Sent Events](#): FastAPI SSE reference the live-chart streaming trick
fastapi.tiangolo.com/reference/sse

6. [SVRPBench](#): NeurIPS 2025 Datasets & Benchmarks, MBZUAI-stochastic traffic modeling
[\(dataset paper\)](#)

Demo Video: [Link](#)
Source Code: [Link](#)